Lecture 25: Markov Chains I

- Markov chains are a type of model for sequences of dependent random variables.
- They have wide applicability.
- This lecture covers basic ideas for Markov chains, including:
  - Random processes
  - State spaces
  - The Markov property
  - Transition probabilities
  - Probability calculations
- Mostly mathematical today; our aim is to illustrate the properties of Markov Chains.

Random Processes

- Consider a sequence of events indexed by time point, $t = 0, 1, 2, \ldots$.
- Let random variable Y_t model outcome at time t.
- Then $\{Y_t : t = 0, 1, 2, \ldots\}$ is a **random process**.
- E.g. Y_t is number of earthquakes of magnitude 4.0 to strike NZ in year t.
- Generalizes to many types of sequences (not only temporal).
  - E.g. Y_t indicates presence (1) or absence (0) of disease in t-th tree along a line transect in a forest.
- We will use temporal terminology for ease of presentation.

Random Processes

Dependence properties

- If earlier events have no impact on process at time t, then $\{Y_t : t = 0, 1, 2, \ldots\}$ is set of independent random variables.
  - Can then model the events using separate statistical models, and ignore the sequential nature.
- But in many real-life applications, value of process at time $t + 1$ will depend on value taken at time t, and possibly earlier.
- E.g. Palmerston North rainfall.
  - Y_t is rain status of day t: 1 if rainy, 0 if dry.
  - Highly plausible that Y_t depends on recent past values.
  - Suggests modelling using conditional probability of Y_t given Y_{t-1}, Y_{t-2} etc.
- Markov chains represent this serial dependence in a convenient, simple fashion.

State Space

- For a random process $\{Y_t : t = 0, 1, 2, \ldots\}$, the **state space** S is the set of values that each Y_t can take.
- Can be discrete or continuous.
  - E.g. $S = \{0, 1\}$ to indicate rainfall status (discrete).
  - E.g. $S = [0, \infty)$ if Y_t is maximum temperature on day t (continuous).
  - We will focus primarily on discrete state space models, unless stated otherwise.
- Common to label states in discrete S using integer values; e.g. $S = \{1, 2, 3, 4, 5, 6\}$.
  - Sometimes relabelling will be convenient; e.g. change $S = \{0, 1\}$ to $S = \{1, 2\}$ for some intents and purposes.

The Markov Property

- There are unlimited possible ways of modelling serial dependence in $\{Y_t : t = 0, 1, 2, \ldots\}$.
- Many produce models that are too complex for easy use.
- The **Markov property** is a powerful simplifying assumption:

$$\mathrm{P}(Y_{t+1} = j | Y_0 = i_0, Y_1 = i_1, \ldots, Y_t = i_t) = \mathrm{P}(Y_{t+1} = j | Y_t = i_t)$$

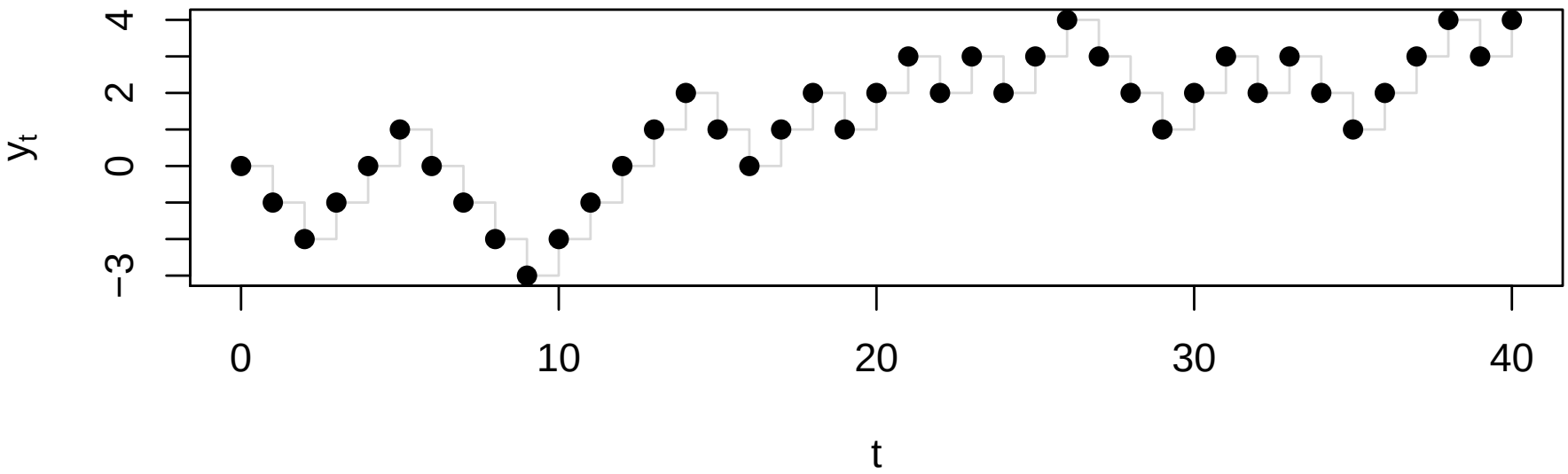
  - $i_0, i_1, \ldots, i_t, j$ are all (time indexed) elements of the state space S
- This indicates that Y_{t+1} is **conditionally independent** of Y_{t-1} (and Y at earlier time points) given Y_t.
- Note this does not imply (unconditional) independence between Y_{t+1} and Y_{t-1}.

Example: A Random Walk on the Integers

- Random walk is an example of a Markov chain.
- If process $\{Y_t : t = 0, 1, 2, \ldots\}$ is in state i at time t, then at time $t + 1$:
  - it will move to state $i + 1$ with some probability ϕ; or
  - or it will move to state $i - 1$ with probability $1 - \phi$.
- Sometimes random walks defined with barriers, so that Y_t cannot go beyond some pre-specified thresholds.

Example continued

Simulation of random walk with $\phi = 1/2$



Example continued

State probabilities at time $t = 41$

- Suppose you want probabilities for the state Y_{41} of the random walk at time $t = 41$.
- Note $Y_{40} = 4$.
- Hence $\mathrm{P}(Y_{41} = 5) = \phi$ and $\mathrm{P}(Y_{41} = 3) = 1 - \phi$.
- Notice these probabilities for Y_{41} do not depend on any of the states prior to time $t = 40$.
  - Could have had completely different course of events over that period, and probabilities for Y_{41} would not change so long as $Y_{40} = 4$ fixed.
  - Illustrates conditional independence of Y_{41} and Y_{39} given Y_{40}.
- But suppose Y_{40} not available.
- Then information from Y_{39} essential in computing probabilities at $t = 41$.

Markov chains

Definition

- A Markov chain is a random process $\{Y_t : t = 0, 1, 2, \ldots\}$ for which the Markov property holds.
- Entire dependence structure specified through the collection of conditional probabilities

$$p_{ij} = \mathrm{P}(Y_{t+1} = j | \ Y_t = i) \qquad\qquad i, j \in S$$

- The probability p_{ij} is called a **transition probability**, since it describes the chance that the Markov chain will transition from state i to state j.
- We consider only cases where these transition probabilities do not depend upon t.
  - Means that our Markov chains are **time-homogeneous**.

Example: Transition probabilities for random walk

For the random walk introduced previously, the transition probabilities are defined by

$$\begin{aligned} p_{i,i+1} &= \phi \\ p_{i,i-1} &= 1-\phi \\ p_{i,j} &= 0 \text{ if } j \text{ does not equal } i-1 \text{ or } i+1. \end{aligned}$$

Transition Matrices

- Transition probabilities conveniently collected together in a **transition matrix** P.
- For Markov chain with states labelled $1, 2, \ldots, m$, P will be an m by m matrix with entry i, j being the transition probability p_{ij}.
- Notice ith row of P contains elements $(p_{i1}, p_{i2}, \ldots, p_{im})$.
  - This is full set of conditional probabilities describing transitions from state i.
  - They must sum to 1: i.e. $\sum_j p_{ij} = 1$.
  - A matrix with rows that sum to one is called **stochastic**.

Example: A Simple Markov chain example

$\{Y_t : t = 0, 1, 2, \ldots\}$ a Markov chain with state space $S = \{1, 2, 3\}$ and transition matrix

$$P = \begin{bmatrix} 0.3 & 0.5 & ? \\ 0.2 & 0.4 & 0.4 \\ 0 & 0.7 & 0.3 \end{bmatrix}$$

Questions

1. What is the missing entry (indicated by the question mark)?
2. If the Markov chain is in state 3 at time t, what is the probability that at time $t + 1$ it will (i) move to state 2; (ii) remain in state 3.

Example: A Simple Markov chain example, continued

Solutions

1. Row probabilities must sum to one, so the missing entry is $1 - 0.3 - 0.5 = 0.2$.
2. The probability of the transition from state 3 to state 2 is $p_{32} = 0.7$. The probability of remaining in state 3 (i.e. making a transition from state 3 to state 3) is $p_{33} = 0.3$.

Another example: Markov chain for Palmy rainfall

- Markov chain $\{Y_t : t = 0, 1, 2, \ldots\}$ models Palmerston North daily rainfall status.
- $Y_t = 1$ if rainy on day t
- $Y_t = 0$ if dry on day t
- Transition probabilities are

$$\begin{aligned} 1 - \alpha &= \Pr(\text{ dry day } t+1 \mid \text{dry day } t) \\ \alpha &= \Pr(\text{ rain day } t+1 \mid \text{dry day } t) \\ \beta &= \Pr(\text{ dry day } t+1 \mid \text{rain day } t) \\ 1 - \beta &= \Pr(\text{ rain day } t+1 \mid \text{rain day } t) \end{aligned}$$

- Can be arranged in a transition matrix

$$P = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$$

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Initial Distribution

- The transition matrix for a Markov chain determines the stochastic behaviour of the chain from time point to time point.
- This does not say anything about distribution of Y_0.
- This is specified by the **initial distribution**.
- Define by row vector $\boldsymbol{\pi}^{(0)}$ of initial probabilities:

$$\boldsymbol{\pi}^{(0)} = \left(\pi_1^{(0)}, \pi_2^{(0)}, \ldots, \pi_m^{(0)}\right)$$

- Hence $\mathrm{P}(Y_0 = i) = \pi_i^{(0)}$.
- We can force Markov chain to start in a specific state by giving initial probability one to this state.
- Knowledge of initial distribution and transition matrix sufficient to compute probabilities (both conditional and unconditional) for state membership at any time point.

Example: some probability calculations

For the rainfall Markov chain

Recall that transition matrix is

$$P = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$$

Suppose that on day $t = 0$, the probability of rain is $\gamma = \mathrm{P}(Y_0 = 1)$. Hence the initial distribution is $\pi^{(0)} = (1-\gamma, \gamma)$.

Questions: Calculate the following probabilities.

1. $\mathrm{P}(Y_1 = 1)$
2. $\mathrm{P}(Y_1 = 1, Y_2 = 0)$
3. Starting on day $t = 0$, what is the probability that of the weather sequence rain, rain, dry, dry dry, rain?
4. If it is raining on Monday, what is the probability that it will also rain on Wednesday?

Rainfall example continued

Solution 1

Q1: Compute $\mathrm{P}(Y_1 = 1)$.

Let A be the event $\{Y_1 = 1\}$ and $B = \{Y_0 = 1\}$, so $B^c = \{Y_0 = 0\}$.

Then $\mathrm{P}(B) = \gamma$ and $\mathrm{P}(B^c) = 1 - \gamma$.

Apply the Law of Total Probability:

$$\begin{aligned}
\Pr(Y_1 = 1) &= \mathrm{P}(A) \\
&= \mathrm{P}(A|B)\mathrm{P}(B) + \mathrm{P}(A|B^c)\mathrm{P}(B^c) \\
&= \mathrm{P}(Y_1 = 1|Y_0 = 1)\mathrm{P}(Y_0 = 1) \\
&\qquad + \mathrm{P}(Y_1 = 1|Y_0 = 0)\mathrm{P}(Y_0 = 0) \\
&= (1 - \beta)\gamma + \alpha(1 - \gamma).
\end{aligned}$$

Rainfall example continued

Solution 2

Q2: Compute $\mathrm{P}(Y_1 = 1, Y_2 = 0)$.

Define events $A = \{Y_2 = 0\}$ and $B = \{Y_1 = 1\}$.

By rules for conditional probabilities:

$$\begin{aligned} \mathrm{P}(Y_1 = 1, Y_2 = 0) &= \mathrm{P}(A \cap B) \\ &= \mathrm{P}(A|\, B)\mathrm{P}(B) \\ &= \mathrm{P}(Y_2 = 0|\, Y_1 = 1)\mathrm{P}(Y_1 = 1) \\ &= \beta[(1-\beta)\gamma + \alpha(1-\gamma)]. \end{aligned}$$

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Rainfall example continued

Solution 3

Q3: Find probability of sequence: rain, rain, dry, dry, dry, rain.

Let day $t = 0$ be Monday, so that $Y_0 = 1$.

Want to find $\mathrm{P}(Y_0 = 1, Y_1 = 1, Y_2 = 0, Y_3 = 0, Y_4 = 0, Y_5 = 1)$.

Factorizes into $\mathrm{P}(Y_0 = 1)\mathrm{P}(Y_1 = 1|Y_0 = 1) \cdots \mathrm{P}(Y_5 = 1|Y_4 = 0)$.

Day, t	0	1	2	3	4	5
Y_t	1	1	0	0	0	1
Probability	γ	p_{11}	p_{10}	p_{00}	p_{00}	p_{01}

Hence probability of the sequence is

$$\gamma \times p_{11} \times p_{10} \times p_{00} \times p_{00} \times p_{01} = \gamma(1-\beta)\beta(1-\alpha)^2\alpha$$

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Rainfall example continued

Solution 4

Q4: If it is raining on Monday, what is the probability that it will also rain on Wednesday?

Let day $t = 0$ be Monday.

Raining on Monday, so $\mathrm{P}(Y_0 = 1) = 1$, and hence $\gamma = 1$.

We now need to compute $\mathrm{P}(Y_2 = 1)$ using this value of γ.

Applying the Law of Total Probability once more,

$$\mathrm{P}(Y_2 = 1) = \mathrm{Pr}(Y_2 = 1|Y_1 = 1)\mathrm{P}(Y_1 = 1) + \mathrm{P}(Y_2 = 1|Y_1 = 0)\mathrm{P}(Y_1 = 0).$$

From part 1, $\mathrm{P}(Y_1 = 1) = 1 - \beta$ so $\mathrm{P}(Y_1 = 0) = 1 - \mathrm{P}(Y_1 = 1) = \beta$. Hence

$$\mathrm{P}(Y_2 = 1) = (1 - \beta)(1 - \beta) + \alpha\beta = (1 - \beta)^2 + \alpha\beta.$$

Probability Calculations with Matrices

Define the distribution at time t by

$$\boldsymbol{\pi}^{(t)} = \left(\pi_1^{(t)}, \pi_2^{(t)}, \ldots, \pi_m^{(t)}\right)$$

- Element $\pi_i^{(t)}$ is probability that chain will be in state i at time t.

It can then be shown that for any time t,

$$\boldsymbol{\pi}^{(t)} P = \boldsymbol{\pi}^{(t+1)}.$$

and so

$$\boldsymbol{\pi}^{(s)} P^t = \boldsymbol{\pi}^{(s+t)}.$$

and in particular

$$\boldsymbol{\pi}^{(0)} P^t = \boldsymbol{\pi}^{(t)}.$$

Follows that P^t is the t-step transition matrix.

Example: Matrix Probability Calculations

For the Rainfall Markov Chain

Repeat the calculations in part (1) and (4) of the previous Example using matrix methods.

Q1: Compute $\mathsf{P}(Y_1 = 1)$.

$$\begin{aligned}
\boldsymbol{\pi}^{(1)} &= \boldsymbol{\pi}^{(0)} P \\
&= (1-\gamma, \gamma) \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} \\
&= ((1-\gamma)(1-\alpha) + \gamma\beta, (1-\gamma)\alpha + \gamma(1-\beta))
\end{aligned}$$

so that $\mathsf{P}(Y_1 = 1) = \pi_1^{(1)} = (1-\gamma)\alpha + \gamma(1-\beta)$ as before.

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Example continued

Q4: If it is raining on Monday, what is the probability that it will also rain on Wednesday?

When $\gamma = 1$, then $\boldsymbol{\pi}^{(0)} = (0, 1)$. Also,

$$
\begin{aligned}
P^2 &= \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} \\
&= \begin{bmatrix} (1-\alpha)^2 + \alpha\beta & (1-\alpha)\alpha + \alpha(1-\beta) \\ \beta(1-\alpha) + (1-\beta)\beta & \beta\alpha + (1-\beta)^2 \end{bmatrix}.
\end{aligned}
$$

Example continued

We can then compute

$$\begin{aligned}
\boldsymbol{\pi}^{(2)} &= \boldsymbol{\pi}^{(0)} P^2 \\
&= (0,1) \begin{bmatrix} (1-\alpha)^2 + \alpha\beta & (1-\alpha)\alpha + \alpha(1-\beta) \\ \beta(1-\alpha) + (1-\beta)\beta & \beta\alpha + (1-\beta)^2 \end{bmatrix} \\
&= \left(\beta(1-\alpha) + (1-\beta)\beta, \beta\alpha + (1-\beta)^2 \right).
\end{aligned}$$

Hence $P(Y_2 = 1) = \pi_1^{(2)} = \beta\alpha + (1-\beta)^2$, as before.

Task/exercise: Hopping Frog



- A frog hops randomly between three lily pads.
- Transition matrix is

$$P = \begin{bmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

- At time $t = 0$, frog is equally likely to be on any lily pad.
- Calculate:
  1. Probability of lily pad sequence $1, 2, 3, 3, 2$.
  2. Probability that frog is on pad 3 at time $t = 2$.

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